Project Report

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Aim: To design an automated irrigation system that uses a soil moisture sensor to control water supply, ensuring efficient water usage and healthy plant growth.

Objectives:

- To design and develop an automated irrigation system using a soil moisture sensor to monitor soil conditions in real time.
- To control the water pump automatically based on the moisture level of the soil, reducing manual intervention.
- To conserve water by supplying it only when required, thereby improving water efficiency.
- To ensure healthy plant growth by maintaining optimum soil moisture levels.

Components Required:

Sno	Name of Component	Qty.
1.	Arduino Uno	1
2.	Relay Module (Single Channel)	1
3.	Soil Moisture Sensor	1
4.	5V Mini Water Pump	1
5.	Jumper Wires	9

Details of Components:

1. **Arduino Uno**

   Arduino Uno is a microcontroller board based on the ATmega328P. It acts as the brain of the system, processing input from the soil moisture sensor and controlling the relay module accordingly. It operates at 5V and can be programmed using the Arduino IDE.

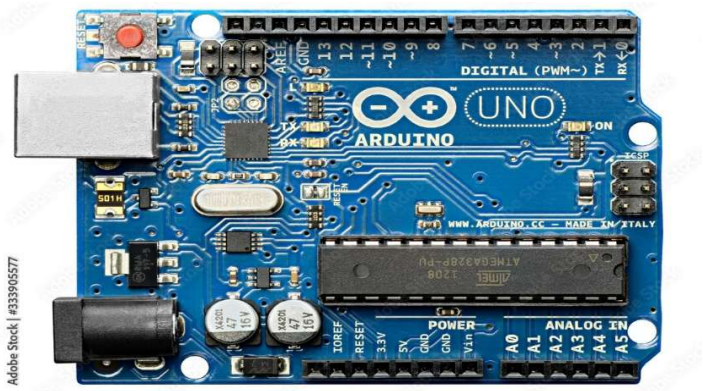

Figure 1 : Arduino Uno

2. **Soil Moisture Sensor:**
   The soil moisture sensor is used to measure the water content in the soil. It works by detecting the electrical conductivity between its probes. When the soil is dry, the resistance is high, and when wet, the resistance decreases. It provides analog output to the Arduino.

3. **Relay Module (Single Channel)**
   The relay module is an electrically operated switch used to control high-power devices like the water pump. It allows the Arduino (low voltage) to safely control the pump (higher power). Most relay modules work on an active LOW signal.

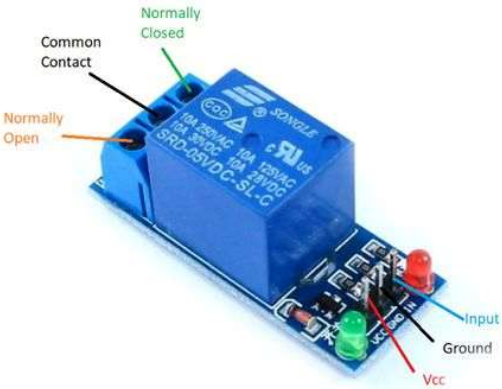

Fig-2(Relay Module)

4. **5V Mini Water Pump**
   The water pump is used to supply water to plants. It gets activated through the relay when the soil moisture level falls below a certain threshold, ensuring automatic irrigation.

5. **Jumper Wires**
   Jumper wires are used to connect different components in the circuit. They help in establishing electrical connections between the Arduino, sensor, and relay module.

Block Diagram of Designed Model:

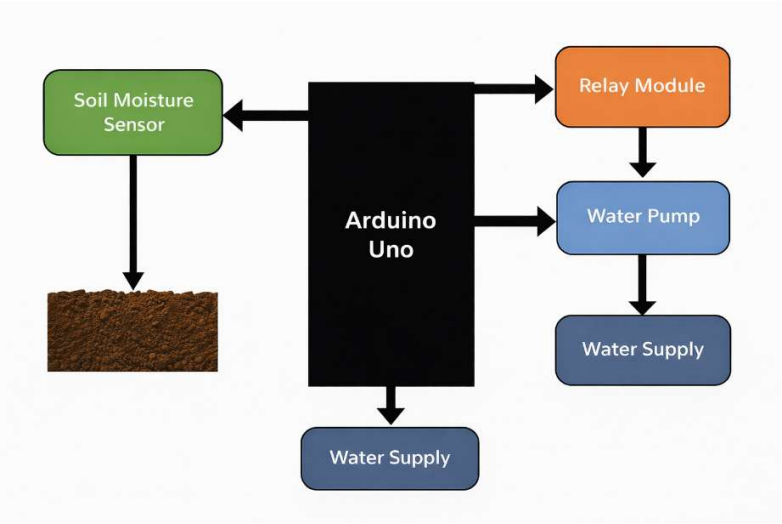

Fig-3(project setup design)

Explanation of Block Diagram :

The designed Smart Irrigation System is an automated system that controls the watering process based on soil moisture conditions. The system mainly consists of a soil moisture sensor, Arduino Uno, relay module, water pump, and water supply.

The soil moisture sensor is placed in the soil to continuously monitor the moisture level. It detects whether the soil is dry or wet and sends the data to the Arduino Uno in the form of analog signals.

The Arduino Uno acts as the central processing unit of the system. It reads the sensor data and compares it with a predefined threshold value. Based on this comparison, it decides whether irrigation is required or not.

If the soil is dry, the Arduino sends a signal to the relay module. The relay module works as an electronic switch and turns ON the water pump. The pump then draws water from the water supply and delivers it to the soil.

Once the soil reaches sufficient moisture level, the Arduino turns OFF the relay, which stops the pump. This process ensures that water is supplied only when needed, preventing water wastage and promoting efficient irrigation.

Working of Designed Model:

As shown in the block diagram, the Smart Irrigation System uses a soil moisture sensor placed in the soil to continuously monitor the moisture level. The sensor detects whether the soil is dry or wet and sends this information to the Arduino Uno.

- The Arduino Uno acts as the central controller of the system. It receives the sensor data and compares it with a predefined threshold value to determine the condition of the soil.
- When the soil moisture level falls below the set threshold (dry condition), the Arduino sends a control signal to the relay module to activate the system.
- The relay module functions as an electronic switch. When it receives the signal from the Arduino, it turns ON the water pump connected to it.
- The water pump starts drawing water from the water supply and delivers it to the soil, ensuring proper irrigation of plants.
- As the soil becomes sufficiently wet, the moisture sensor detects the change and sends updated data to the Arduino Uno.
- Once the moisture level reaches the desired limit, the Arduino turns OFF the relay module, which stops the water pump.
- This automatic process continues, ensuring efficient water usage, reducing manual effort, and maintaining optimal soil conditions for plant growth.

Source Code:

```
const int sensorPin = A0;
const int relayPin = 7;

int sensorValue = 0;
int lastValue = -1;
bool pumpState = false;

void setup() {
  Serial.begin(9600);
  pinMode(relayPin, OUTPUT);
  digitalWrite(relayPin, HIGH);
}
void loop() {
  sensorValue = analogRead(sensorPin);
  if(abs(sensorValue - lastValue) > 10)
  {
    Serial.print("Soil Value: ");
    Serial.println(sensorValue);
    lastValue = sensorValue;
  }

  if(sensorValue > 600)
  {
    if(pumpState == false) {
      digitalWrite(relayPin, LOW);
      Serial.println("Pump ON");
      pumpState = true;
    }
  }
  else
  {
    if(pumpState == true) {
      digitalWrite(relayPin, HIGH);
      Serial.println("Pump OFF");
      pumpState = false;
    }
  }
```

```
  delay(1000);
}
```

Pictures of Prototype:

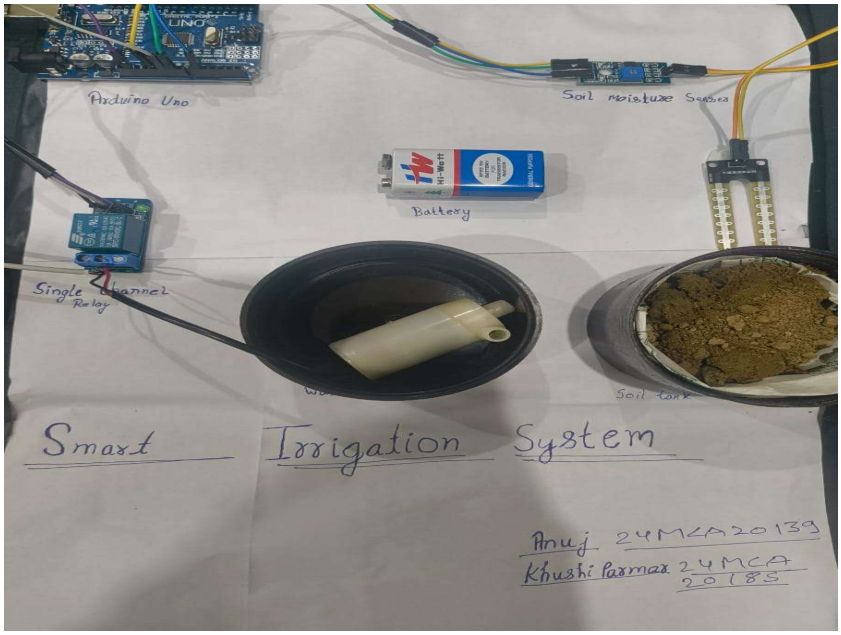

Figure 4 : smart irrigation system

Learning outcomes (What I have learnt):

1. Understand the concept of automated irrigation systems and their importance in water conservation.
2. Gain knowledge about interfacing a soil moisture sensor with the Arduino Uno for real-time data monitoring.
3. Learn how to control electrical devices like a water pump using a relay module.
4. Develop programming skills using the Arduino IDE for sensor-based automation.
5. Analyze sensor data and implement decision-making logic based on threshold values.